CRYPTOLOGY - 5

* Rail fence:-

1) encrypt the message "ENJOY YOUR JOURNEY" using Rail Fence with depth 2 and 3

⇒ ENJOY YOUR JOURNEY

depth = 2

E J Y O R O R E
N O Y U J U N Y

= EJYORORENOYUJUNY

depth = 3

E Y R R
N O Y U J U N Y
J O U E

= EYRRNOYUJUNYJOUE

2) encrypt the message as per your, your choice using Rail Fence with depth 2 ~~or~~ ~~or~~ and 3.

⇒ depth = 2

CRYPTOLOGY

C Y T L G
R P O O Y

⇒ CYTLGRPOOY

depth=3

CRYPTOLOGY

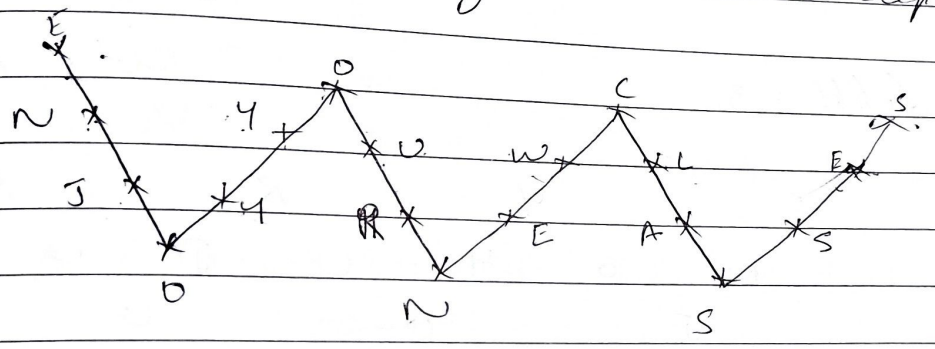
C T G

R P O O Y

Y L Y

CTGRPOOYYLX

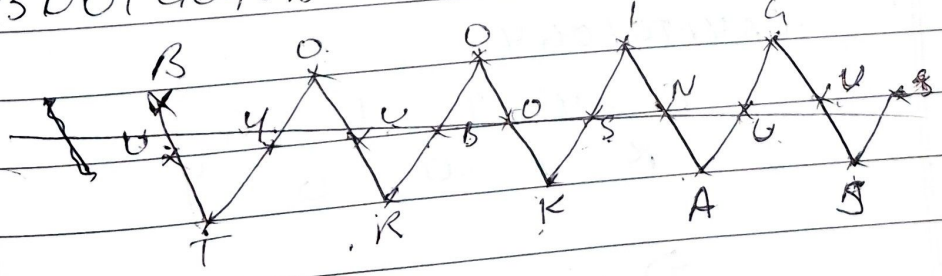
③ decrypt the ciphur "EOLSNUYUWLEJYREASONS" encrypted using Railfence with depth 4



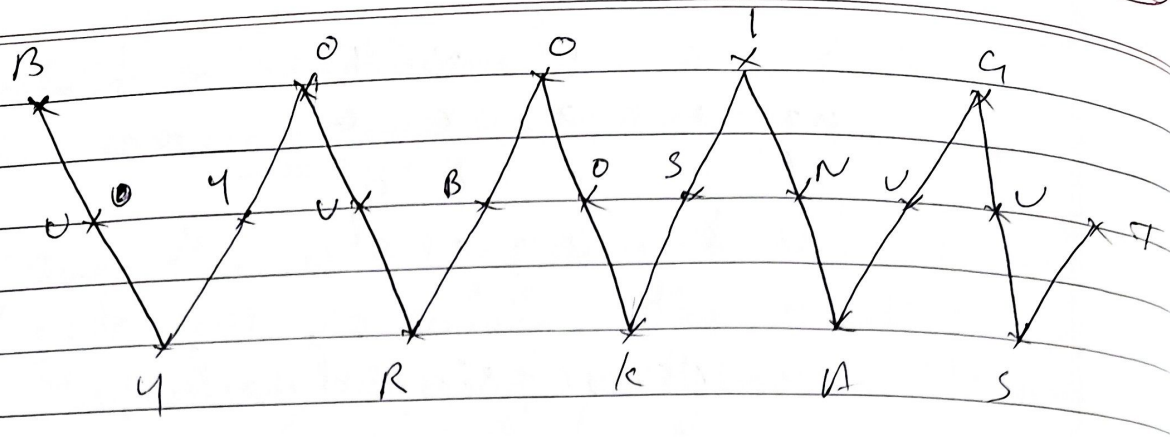
= ENJOY YOUR NEW CLASSES

④ decrypt the ciphur "BOOIGUYUBOSNUUTYRKEAS" encrypted using Railfence with depth 2

BOOIGUYUBOSNUUTRKEAS



= BUY YOUR BOOKS IN AUGUST



= BUY YOUR BOOKS IN AUGUST

⑤ Decrypt the cipher "SEMSERTISINCMOO" encrypted using Rail fence with depth 2.

SEMSERTISINCMOO

There are 15 letters
- first row → 8 letters
- second row → 7 letters

SEMSERTI|SINCMOO

SSEIMNSCEMROTOI

2) Permutation:-

① encrypt the following message "we are discovered" using the key "53124".

key : 5 3 1 2 4

encrypt : we are discovered

5 3 1 2 4

w e a r e

d i s c o

v e r e d

Answer: discovered

③ DES

① using the initial permutation table, determine the output of the initial permutation box when the input is given in hexadecimal as:

0x0002 0000 0000 0001

⇒ It takes 64 bits as input

0000200000000001

hexadecimal digit = 4 bits, so this is 64 bits

binary conversion

Hexadecimal: 0x0002 0000 0000 0001

Binary: 0000 0000 0000 0010 } Input to box (plain text)
0000 0000 0000 0000
0000 0000 0000 0000
0000 0000 0000 0001

10th

0000 0000 0000 0010 0000 0000 0000 0000
0000 0000 0000 0000 0000 0000 0000 0010

64th

Input bit 64 = 1, output position 25 = 1
Input bit 15 = 1, output position 63 = 1
} By using Permutation table

64-bit output:

0000000 0000000 0000000 0000000 1000000 0000000 0000000 0000000
0000000 0000000 0000000 0000010

output in hexadecimal:

0x00000008 00000000₂

2) Given the 4-bit plaintext (M) 1101 and key (k) 1011, calculate the resulting ciphertext (C), using a bitwise exclusive-OR (XOR) operation. Show your bit-by-bit work.

XOR		
A	B	A⊕B
0	0	0
0	1	1
1	0	1
1	1	0

⇒ plaintext: 1101

key: 1011

```
  1101
  1011
  ----
  0110
```

ciphertext: 0110

3) The input to s-box 1 is 000000. what is the output.

⇒ 000000

1st bit, 6th bit — 1st and 6th → row

middle 4 bits → column

So, 0th row and 0th column see the number in s-box → 14

Binary conversion of 14 → 1110

output = 1110